

JEE MAIN REVISION MASTER: D AND F BLOCK ELEMENTS 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet D And F Block Elements - Core Application	28 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet D And F Block Elements - Core Application (Very High Yield — 28 PYQs)

Core Principle: Key problem pattern in Kcet D And F Block Elements.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The calculated spin only magnetic moment of Cr^{2+} ion is..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "The highest oxidation state of manganese in fluoride is +4 (MnF_4), but the highest oxidation state in oxides is +7 (Mn_2O_7)" [Ans: D]
- [2026 Shift] Q#3 (MCQ): "Which of the following will not act as an oxidising agent?" [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The calculated spin only magnetic moment of Cr^{2+} ion is

- (A) 3.87 BM
- (B) 4.90 BM **[Correct]**
- (C) 5.92 BM
- (D) 2.84 BM

Correct Answer: B

Solution Analysis: For spin only magnetic moment, we use the formula $\mu = \sqrt{n(n+2)}\text{BM}$ where n is the number of unpaired electrons. Chromium has atomic number 24. Its electronic configuration is $\text{Cr} = [\text{Ar}]3d^5 4s^1$. For Cr^{2+} , two electrons are removed. First the 4s electron is removed, then one electron from 3d. So, $\text{Cr}^{2+} = [\text{Ar}]3d^4$. In $3d^4$, according to Hund's rule, the four electrons occupy four different d orbitals singly. So the number of unpaired electrons is $n=4$. Now, $\mu = \sqrt{4(4+2)} = \sqrt{24}$ $\mu = 4.90\text{BM}$. So the co